Theory Note 1 of 8

TCP Three-Way Handshake and Connection Teardown

Network Traffic Analysis, Nmap and Windows Logging | Week 2

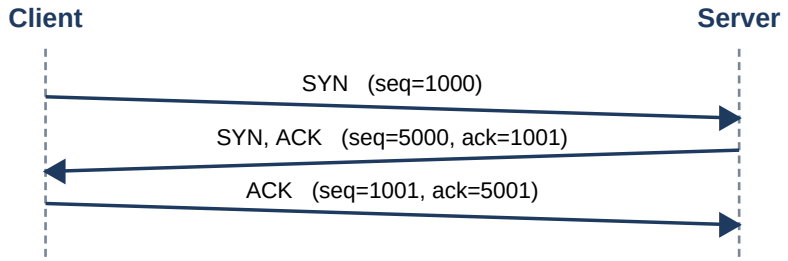
1. Overview

TCP is a connection-oriented protocol. Before two hosts exchange any application data, they first agree that a connection exists and settle the starting sequence numbers each side will use. That agreement is the three-way handshake. When the conversation is over, a separate exchange closes the connection cleanly. For a SOC analyst these two phases are worth knowing well, because the packets that open and close a connection show quickly whether a session was genuine, failed, or was only a probe.

2. The three-way handshake

The handshake takes three packets. The client starts it, the server answers, and the client confirms.

- SYN: the client sends a segment with the SYN flag set and an initial sequence number (ISN) it has picked, for example 1000. In effect it is asking, "Can we talk?"
- SYN-ACK: if the port is open, the server replies with both SYN and ACK set. It sends its own ISN (say 5000) and acknowledges the client's with ack=1001, which is the client's ISN plus one.
- ACK: the client acknowledges the server's number with ack=5001. Both sides are now in the ESTABLISHED state and data can flow.



Step	Direction	Flags set	Example numbers	State afterwards
1	Client to server	SYN	seq=1000	Client: SYN_SENT
2	Server to client	SYN, ACK	seq=5000, ack=1001	Server: SYN_RECEIVED
3	Client to server	ACK	seq=1001, ack=5001	Both: ESTABLISHED

Three steps are needed because each side has to announce its own sequence number and have it acknowledged by the other. Two packets would leave one direction unconfirmed. Random starting numbers also make it harder for an outsider to inject packets into a session or confuse it with an old, delayed one.

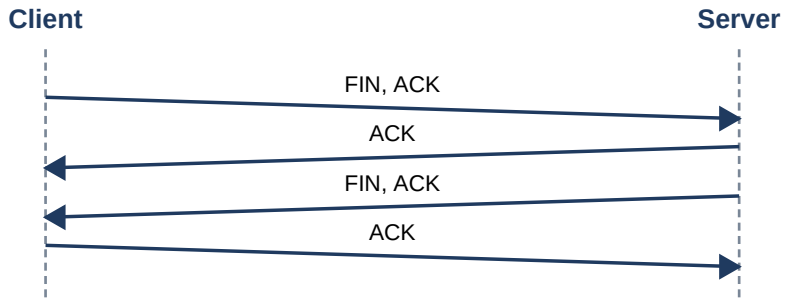
Not every attempt ends in a full handshake. How the server reacts to the first SYN tells you the state of the port:

Server response to SYN	Meaning	How Nmap reports it
SYN-ACK	Port is open and a service is listening.	open
RST (or RST, ACK)	Port is closed; the host is alive but nothing is listening.	closed
No reply, or ICMP unreachable	A firewall is dropping the packet, or the host is down.	filtered

3. Connection teardown

TCP closes each direction of a connection on its own, so a graceful close normally takes four packets.

- Side A has finished sending and transmits FIN, ACK.
- Side B acknowledges it with ACK. The A-to-B direction is closed, but B may still send data.
- When B is also finished, it sends its own FIN, ACK.
- Side A answers with a final ACK and waits in the TIME_WAIT state, usually for a minute or two, so late packets from the old connection are not mistaken for a new one.



In practice, steps 2 and 3 are often combined into a single FIN, ACK packet, so only three packets may appear on the wire. A connection can also end abruptly: if either side sends a RST, the connection is dropped at once with no further acknowledgement. This happens when a packet reaches a closed port, when an application crashes, when a firewall or IPS intervenes, or when a scanner such as Nmap abandons a half-open connection.

4. TCP flags

Flag	Full name	What it does	What an analyst should notice
SYN	Synchronize	Starts a connection and carries the initial sequence number.	Many SYNs from one source with no completed handshakes point to a port scan or SYN flood.
ACK	Acknowledgement	Confirms received data. Set on nearly every packet after the first SYN.	ACK-only probes can be used to map firewall rules (Nmap -sA).
FIN	Finish	Says there is no more data to send and starts a graceful close.	A FIN with no existing session suggests a FIN scan or an evasion attempt.
RST	Reset	Aborts the connection immediately.	A burst of RSTs returning to one host usually means a scanner is hitting closed ports.
PSH	Push	Asks the receiver to hand data to the application straight away instead of buffering it.	Normal in HTTP and SSH; unusual on packets that carry no data.
URG	Urgent	Marks the urgent pointer as valid so flagged data is handled first.	Rarely used legitimately. FIN, PSH and URG together form the Xmas scan.

> **Key takeaway**
> A full SYN, SYN-ACK, ACK sequence proves a connection was truly established, while SYN followed by RST shows it was only probed. The filter tcp.flags.syn==1 && tcp.flags.ack==0 isolates the start of every connection attempt and is the pattern seen in a SYN scan.